Matematisk statistik
Matematikcentrum / Lunds universitet

Resit exam: 2023–11–24 kl 14^{00}–19^{00}
MASA02 ÄMAD04 Probability theory

All solutions should be well justified. The total amount of points is 90; to pass, you need to get at least 40. Use the department's papers only. Begin the solution of each assignment **on top of a new sheet**. Do **not** use **red** pen or pencil. You are allowed to use a pocket calculator.

Write your identifier on each sheet.

1. State the axioms of probability. Rigorously, from the axioms, prove that $\mathbb{P}(\emptyset) = 0$ and that $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ whenever $A \cap B = \emptyset$. (10p)

2. If a student prepares for the exam, then s/he passes it with probability α. If s/he does not prepare, s/he passes it with probability $\alpha/2$. Suppose that a randomly chosen student prepares for the exam with probability α and does not prepare with the complementary probability. We know that of a large group of students, each behaving independently, $1/8$ passed the exam. What is the value of α? (10p)

3. Suppose (X, Y) are jointly continuous with the density $f_{X,Y}(x, y) = c$ for (x, y) inside of the unit circle $x^2 + y^2 \leq 1$, while $f_{X,Y}(x, y) = 0$ for other values. (20p)

   What is the value of c? Compute the *conditional* density of random variable X given $Y = y$ as well as $\mathbb{E}(X \mid Y)$ and $\mathsf{Var}(X \mid Y)$.

4. Give the definition of convergence in $\mathbb{L}^p$, $p > 0$, and the definition of convergence in probability. Suppose that $X_n \to X$ in $\mathbb{L}^p$; rigorously prove that $X_n \to X$ in probability. (20p)

   Give an example of a sequence of random variables X_n such that $X_n \to X$ in probability and in $\mathbb{L}^1$, but not in $\mathbb{L}^2$.

5. At each round, you roll two dice, simultaneously and independently, and get the numbers X_n and Y_n (both in $\{1, 2, 3, 4, 5, 6\}$) respectively, $n = 1, 2, \ldots$. You stop when the difference of the numbers the two dice show is no more than 1 in absolute value, i.e., $|X_n - Y_n| \leq 1$. What is the distribution of the random variable $Z \in \{1, 2, \ldots\}$, the number of rounds you require? (20p)

   Find $\mathbb{P}(X_1 = k \mid Z = 2)$ for $k = 1, 2, 3, 4, 5, 6$ and then compute $\mathbb{E}(X_1 \mid Z = 2)$.

6. State and rigorously (from the axioms of probability) prove the "continuity of probability theorem". (10p)

   Let F_X be a cumulative distribution function of some random variable X. Prove that $F_X(\cdot)$ is right-continuous, i.e., $\lim_{x \downarrow a} F_X(x) = F_X(a)$. Also prove that $\lim_{x \to +\infty} F_X(x) = 1$.

Good luck!

Solutions

1. Axioms: see lectures. For the second part, see also lecture notes, or just follow this argument. Let $A_1, A_2, \ldots$ all equal $\emptyset$. Then $A = \cup_n A_n = \emptyset$ and A_i's are disjoint as any pairwise intersection is $\emptyset$. Let $x = \mathbb{P}(\emptyset)$. By Axiom 3, $\mathbb{P}(A) = \sum_n \mathbb{P}(A_n)$, i.e., $x = \sum_{n=1}^{\infty} x$. This is only possible if $x = 0$.

   For the other part, set $A_1 = A$, $A_2 = B$, $A_3 = A_4 = \cdots = \emptyset$ and apply Axiom 3 to the sequence $\{A_n\}$.

2. Let $A =$"passes", $B =$"prepares". By the law of total probability, $\mathbb{P}(A) = \mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c)$, i.e. $\alpha \cdot \alpha + \alpha/2 \cdot (1-\alpha) = 1/8 \iff -4\alpha^2 + 4\alpha = 1$ i.e. $\alpha = (\sqrt{2}-1)/2 \approx 0.2071$.

3. $\iint f(x,y)\mathrm{d}x\mathrm{d}y = 1$ hence $c = \frac{1}{\pi}$. Given $Y = y$, the random variable X is Uniform on $[-\sqrt{1-y^2}, +\sqrt{1-y^2}]$. For $U \sim U[-a,a]$ we have $\mathbb{E}U = 0$ and $\mathsf{Var}U = a^2/3$. Here $a = \sqrt{1-y^2}$.

4. For the definitions, see lectures. If $X_n \to X$ in L^p, then for any $\varepsilon > 0$ we have $\mathbb{P}(|X_n - X| \geq \varepsilon) = \mathbb{P}(|X_n - X|^p \geq \varepsilon^p) \leq \mathbb{E}(|X_n - X|^p)/\varepsilon^p \to 0$ by Markov's inequality, so $X_n \to X$ in probability.

   Example: $\mathbb{P}(X_n = n) = 1/n^2 = 1 - \mathbb{P}(X_n = 0)$, $X = 0$. Then $\mathbb{E}|X_n - X| = n \cdot \frac{1}{n^2} \to 0$ but $\mathbb{E}(|X_n - X|^2) = n^2 \cdot \frac{1}{n^2} \equiv 1 \not\to 0$.

5. $\mathbb{P}(|X_n - Y_n| \leq 1) = \sum_{k=1}^{6} \mathbb{P}(|X_n - Y_n| \leq 1 \mid Y_n = k)\mathbb{P}(Y_n = k) = \left[\frac{2}{6} + \frac{3}{6} + \frac{3}{6} + \frac{3}{6} + \frac{3}{6} + \frac{2}{6}\right] \frac{1}{6} = \frac{16}{36} = \frac{4}{9}$ so $Z = 1, 2, \ldots$ is Geometric(p) with $p = 4/9$; $\mathbb{P}(Z = k) = (1-p)^{k-1}p$, $k = 1, 2, \ldots$.

   $\mathbb{P}(X_1 = k \mid Z = 2) = \frac{\mathbb{P}(X_1 = k, Z = 2)}{\mathbb{P}(Z=2)}$ and

$$\begin{aligned}
\mathbb{P}(X_1 = k, Z = 2) &= \mathbb{P}(X_1 = k, |Y_1 - k| > 1, |X_2 - Y_2| \leq 1) \\
&= \mathbb{P}(X_1 = k) \cdot \mathbb{P}(|Y_1 - k| > 1) \cdot \mathbb{P}(|X_2 - Y_2| \leq 1) \\
&= \frac{1}{6} \cdot \mathbb{P}(|Y_1 - k| > 1) \cdot \frac{4}{9}
\end{aligned}$$

   because of independence. Now, $\mathbb{P}(|Y_1 - k| > 1)$ equals $\frac{4}{6}$ if $k = 1, 6$ and $\frac{3}{6}$ if $k = 2, 3, 4, 5$ so $\mathbb{P}(X_1 = k \mid Z = 2) = \begin{cases} 1/5 & k = 1, 6; \\ 3/20 & k = 2, 3, 4, 5 \end{cases}$
   and $\mathbb{E}(X_1 \mid Z = 2) = (1+6) \times \frac{1}{5} + (2+3+4+5) \times \frac{3}{20} = 3.5$

   Note that X_1 and Z are *not* independent, but X_1 is symmetric w.r.to $k \to 7-k$ even given $Z = 2$, hence it's not surprsing that $\mathbb{E}(X_1 \mid Z = 2) = 7/2$.

6. For the proofs, please see the lecture notes.